

ADVANCED MACHINE LEARNING · WORKSHEET 07

Regression and Classification
Evaluation MetricsResiduals · MAE · MAPE · MSE · RMSE · R^2 · Adjusted R^2 · Confusion Matrix · Precision · Recall · F1

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▷ CENTRAL QUESTION

How do we know whether a trained model is actually good? A useful metric must match the **kind and cost of error** that matters for the decision.

PART 1
REGRESSION EVALUATION METRICS

SECTION 1 Training Has Ended; Evaluation Begins

▷ CENTRAL QUESTION

The model has already chosen its parameters. Are we measuring how it *learns*, or judging how well the finished model performs?

	Training loss	Evaluation metric
Main role	Guides parameter updates during training	Judges a fixed model after training
Typical data	Training data	Validation or test data for generalisation
Example	MSE used by an optimiser	MSE, MAE, R^2 , F1, etc. reported after fitting

★ KEY INSIGHT

The same expression can play both roles. MSE is a **loss** when it guides learning and an **evaluation metric** when it judges a finished model. Context determines the role.

CHECKPOINT 1 · SET THE CONTEXT

For each statement, write **L** for training loss or **E** for evaluation metric.

(a) It is calculated on a held-out test set after parameters are fixed. **E**

(b) The optimiser uses it to update coefficients. **L**

(c) It answers whether the finished model is acceptable for a decision. **E**

Can one number tell us everything about a model? Explain briefly.

Answer: No. Different metrics answer different questions about typical errors, large errors, relative errors, baselines, false alarms, missed classes, and class imbalance.

SECTION 2 One Regression Test Dataset and Signed Residuals

▷ CENTRAL QUESTION

Model A predicts monthly internship stipend in **thousands of INR**. For one student, did the model predict too high or too low, and by how much?

Signed residual (provided):

$$e_{A,i} = y_i - \hat{y}_{A,i}$$

A positive residual means Model A predicted too low; a negative residual means it predicted too high. A residual preserves direction; it does not yet summarise the whole model.

MASTER TEST DATA R · USE THIS ONE TABLE THROUGHOUT PART 1

The actual stipends and both models' predictions are provided. S1 is worked; complete the remaining derived cells in Homework H1. Every regression calculation in Part 1 uses this frozen test dataset.

ID	y_i	$\hat{y}_{A,i}$	$\hat{y}_{B,i}$	$e_{A,i}$	$ e_{A,i} $	$e_{A,i}^2$	APE _A (%)	$e_{B,i}^2$
S1	30	32	32	-2	2	4	6.67	4
S2	35	34	34	+1	1	1	2.86	1
S3	40	43	43	-3	3	9	7.50	9
S4	45	42	43	+3	3	9	6.67	4
S5	50	49	49	+1	1	1	2.00	1
S6	55	58	58	-3	3	9	5.45	9
S7	60	56	56	+4	4	16	6.67	16
S8	65	70	69	-5	5	25	7.69	16
Total	-	-	-	-4	22	74	45.50	60

Column guide:

$$|e_{A,i}| = |y_i - \hat{y}_{A,i}|, \quad \text{APE}_{A,i} = 100 \left| \frac{e_{A,i}}{y_i} \right|, \quad e_{B,i}^2 = (y_i - \hat{y}_{B,i})^2.$$

The residual, absolute-error, squared-error and APE columns refer to **Model A**. Model B and $e_{B,i}^2$ are retained for the Adjusted R² comparison in Section 5.

Worked row S1: $e_{A,1} = 30 - 32 = -2$. Model A overpredicted S1's stipend by 2 thousand INR.

Rounding note: the APE total uses unrounded fractions. The displayed row percentages are rounded.

SECTION 3 Regression Error Metrics

▷ CENTRAL QUESTION

Signed residuals show direction, but positive and negative values can cancel. How should we summarise the size of all eight errors?

3.1 Mean Absolute Error (MAE)

Question	What is the typical absolute miss in the original target unit?
Formula and result	$\text{MAE} = \frac{1}{n} \sum_{i=1}^n e_{A,i} = \frac{22}{8} = 2.75$ thousand INR.
Advantage	Easy to interpret; every extra unit of error receives equal linear weight.
Disadvantages	Hides direction, depends on target scale, and does not emphasise a rare large miss.

3.2 Mean Absolute Percentage Error (MAPE)

Question	How large is the typical error relative to the actual value?
Formula and result	$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left \frac{e_{A,i}}{y_i} \right = \frac{45.50\%}{8} \approx 5.69\%.$
Advantage	A scale-free percentage that is convenient when relative error matters.
Disadvantages	Undefined at $y_i = 0$, unstable near zero, and can overemphasise small actual values.

3.3 Mean Squared Error (MSE) and Root Mean Squared Error (RMSE)

Metric	Formula and Model A result	Advantage	Disadvantage
MSE	$\frac{1}{n} \sum_{i=1}^n e_{A,i}^2 = \frac{74}{8} = 9.25$ (thousand INR) ²	Strongly penalises large misses; useful in optimisation.	Squared units; sensitive to unusual large errors.
RMSE	$\sqrt{\text{MSE}} = \sqrt{9.25} \approx 3.04$ thousand INR	Keeps the large-error penalty but returns to the target unit.	Still scale-dependent and more outlier-sensitive than MAE.

PRACTICE P2 · APPLY THE METRIC MAP

Match each decision to the most useful metric.

Decision need	Metric
1 Typical miss in stipend units; equal linear weight per unit	MAE
2 Compare relative error across scales; actuals stay away from zero	MAPE
3 Give large misses extra influence and report in stipend units	RMSE
4 Use a squared penalty during optimisation	MSE

Concept check. If one absolute residual rises from 1 to 5, its contribution to $\sum |e_i|$ rises by 4, while its contribution to $\sum e_i^2$ rises by 24. Which metric family reacts more strongly? **MSE / RMSE**

True or false. “MSE = 9.25 means a typical miss of 9.25 thousand INR.” **False** MSE has squared units; RMSE ≈ 3.04 thousand INR is the comparable original-unit summary.

HOMEWORK H1 · ERROR-SIZE METRICS FROM THE MASTER TABLE

Show your working on a separate sheet.

- Complete the unfilled derived cells in Master Test Data R.
- Verify $\sum |e_A| = 22$, $\sum e_A^2 = 74$, $\sum \text{APE}_A \approx 45.50\%$, and $\sum e_B^2 = 60$; then reproduce all four results above.
- Explain when you would report MAE, MAPE, or RMSE, and state MAPE's zero-value limitation.

Teacher key: MAE = 2.75 thousand INR; MAPE $\approx 5.69\%$; MSE = 9.25 (thousand INR)²; RMSE ≈ 3.04 thousand INR.

SECTION 4 R² from the Mean-Only Baseline

▷ CENTRAL QUESTION

The four previous metrics tell us how large the residuals are. Is Model A meaningfully better than a model that predicts the same stipend for everyone?

Step 1 - Build the mean-only baseline.

$$\bar{y} = \frac{30 + 35 + 40 + 45 + 50 + 55 + 60 + 65}{8} = \underline{47.5}$$

The baseline predicts 47.5 thousand INR for every student.

Step 2 - Measure the total variation available to explain.

$$\begin{aligned} \text{TSS} &= \sum_{i=1}^n (y_i - \bar{y})^2 \\ &= 306.25 + 156.25 + 56.25 + 6.25 + 6.25 + 56.25 + 156.25 + 306.25 = \underline{1050} \end{aligned}$$

★ KEY INSIGHT

TSS is the total squared variation of actual stipends around their mean. It is the variation the trained model has the opportunity to account for. If we predict only the mean, all 1050 remains unexplained.

Step 3 - Measure the variation left after using Model A.

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \underline{74}$$

RSS is the residual sum of squares: the squared variation still unexplained by Model A.

Step 4 - Compare leftover variation with original variation.

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}} = 1 - \frac{74}{1050} = \underline{0.9295}$$

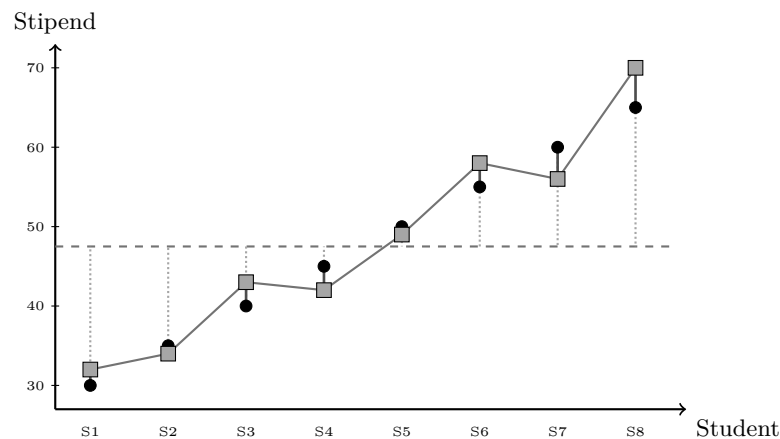
Equivalent explained-fraction view:

$$\frac{\text{TSS} - \text{RSS}}{\text{TSS}} = \frac{976}{1050} = \underline{0.9295}$$

★ KEY INSIGHT

On these students, Model A accounts for about **92.95% of the squared variation** that existed around the mean-only baseline. About 7.05% remains unexplained.

Visual: actual stipends, Model A, and the mean-only baseline



Legend: black dots = actual; gray squares/solid line = Model A; dashed line = mean-only baseline.

Each dotted actual-to-mean gap contributes to **TSS**; each solid actual-to-square gap contributes to **RSS**. The global explained variation is **TSS - RSS = 1050 - 74 = 976**; it is not a third set of point-to-point gaps.

Interpret the landmarks.

Value	Correct interpretation
$R^2 = 1$	Perfect fit: predictions match every actual value and $\text{RSS} = 0$.
$R^2 = 0$	No improvement over the mean-only baseline: $\text{RSS} = \text{TSS}$.
$0 < R^2 < 1$	Better than the mean-only baseline, but some variation remains unexplained.
$R^2 < 0$	Worse than the mean-only baseline on the evaluated data: $\text{RSS} > \text{TSS}$.

DO NOT LET THIS MISCONCEPTION SURVIVE

$R^2 = 0.93$ does not mean 93% accuracy, 93% of predictions are correct, or a 7% average error. R^2 is a relative sum-of-squares comparison with the mean baseline. Report MAE or RMSE for error size.

PRACTICE P3 · SAME RMSE, DIFFERENT R^2

Two datasets can have the same RMSE but different R^2 values. What additional property changes R^2 ?

Answer: The spread of the actual target values changes TSS. With the same RSS/RMSE, a wider target spread produces a larger TSS and can make the same residual error look better in relative terms.

SECTION 5 R^2 Limitations and Adjusted R^2 **▷ CENTRAL QUESTION**

If adding the last digit of a student's roll number makes training R^2 rise slightly, has that extra predictor earned a place in the model?

For ordinary least squares with an intercept on the training data, adding a predictor cannot increase RSS; therefore training R^2 cannot decrease. Even a useless feature may fit noise. R^2 rewards fit but does not charge for complexity.

Adjusted R^2 formula (provided):

$$R_{\text{adj}}^2 = 1 - (1 - R^2) \frac{n - 1}{n - p - 1}$$

where n is the number of observations and p is the number of predictors, excluding the intercept. The factor $(n - 1)/(n - p - 1)$ grows as p grows, so a new predictor must reduce unexplained variation enough to overcome the larger penalty.

★ ADJUSTED R^2 : TERM-BY-TERM

- R^2 is the model's raw improvement over the mean-only baseline; $1 - R^2$ is the fraction left unexplained.
- n is the number of observations. The numerator $n - 1$ is tied to estimating the target mean.
- p is the number of predictors, excluding the intercept; $n - p - 1$ is the residual degrees of freedom.
- The ratio $(n - 1)/(n - p - 1)$ enlarges the unexplained fraction as predictors are added. The outer 1- converts it back to an adjusted explained-variation score.

Model B predicts (32, 34, 43, 43, 49, 58, 56, 69) for S1–S8; its extra predictor reduces RSS from 74 to 60.

Model	Predictors	p	RSS	R^2	Adjusted R^2
A	CGPA + interview score	2	74	0.9295	0.9013
B	Model A + roll-number last digit	3	60	0.9429	0.9000

WORKED COMPARISON · DID THE EXTRA FEATURE EARN ITS PLACE?

For $n = 8$, the completed calculations are provided.

$$R_{\text{adj},A}^2 = 1 - \left(\frac{74}{1050} \right) \frac{7}{5} = 0.9013$$

$$R_{\text{adj},B}^2 = 1 - \left(\frac{60}{1050} \right) \frac{7}{4} = 0.9000$$

Here $1 - R^2 = \text{RSS}/\text{TSS}$, so these calculations avoid carrying rounded R^2 values. Model B has the higher raw R^2 , but Model A has the higher Adjusted R^2 .

Decision: The roll-number feature did not earn its place. Model B reduced RSS and raised raw R^2 , but the improvement was too small to offset the third-predictor penalty.

★ WHAT ADJUSTED R^2 DOES - AND DOES NOT - TELL US

- Useful for comparing regression models fitted to the same target and observations with different p .
- Can decrease when a new feature adds too little explanatory value; normally $R^2_{\text{adj}} \leq R^2$.
- Not an automatic feature-selection oracle and not a replacement for validation or cross-validation.
- Requires $n > p + 1$, and p does not include the intercept.

◆ TAKEAWAY

Signed residual $\rightarrow$ absolute size $\rightarrow$ relative size $\rightarrow$ large-error penalty $\rightarrow$ target units $\rightarrow$ mean baseline $\rightarrow$ feature-count penalty.

PART 2 CLASSIFICATION EVALUATION METRICS

SECTION 6 Bridge: From Error Size to Error Identity

▷ CENTRAL QUESTION

In regression, predicting 50 instead of 55 creates a five-unit error. If a classifier predicts the wrong label, what is the numerical distance between “Ready” and “Needs Practice”?

	Regression	Classification
Model output	A number, such as stipend	A label, such as Ready
First error question	How large is the error?	Which label was confused with which?
First object to inspect	Signed residual	Actual–predicted label pair

The placement team now uses a second model to assign one of three internship-readiness bands: **Ready (R)**, **Needs Practice (P)**, or **Needs Support (S)**. We begin with the simplest classification summary: count exact matches.

◆ TAKEAWAY

Regression errors have **size**. Classification errors have **identity**. The next question is which class was confused with which other class.

SECTION 7 Multiclass Accuracy and the Confusion Matrix

▷ CENTRAL QUESTION

The classifier is 66.7% accurate. What structure is hidden inside that single number?

MASTER TEST DATA C · USE THIS ONE TABLE THROUGHOUT PART 2

This is the frozen classification test dataset. Every accuracy, confusion-matrix, Needs-Support, and aggregate calculation in Part 2 comes from this table. The Result column is provided so the activity can focus on organising and analysing the errors.

Needs Support is the smallest class (4 students), but it is operationally important because these students may need timely intervention.

Student	Actual class	Predicted class	Result
S01	R - Ready	R - Ready	Correct
S02	R - Ready	R - Ready	Correct
S03	R - Ready	R - Ready	Correct
S04	R - Ready	R - Ready	Correct
S05	R - Ready	R - Ready	Correct
S06	R - Ready	R - Ready	Correct
S07	R - Ready	P - Needs Practice	Error
S08	R - Ready	S - Needs Support	Error
S09	P - Needs Practice	R - Ready	Error
S10	P - Needs Practice	P - Needs Practice	Correct
S11	P - Needs Practice	P - Needs Practice	Correct
S12	P - Needs Practice	P - Needs Practice	Correct
S13	P - Needs Practice	P - Needs Practice	Correct
S14	P - Needs Practice	S - Needs Support	Error
S15	S - Needs Support	P - Needs Practice	Error
S16	S - Needs Support	P - Needs Practice	Error
S17	S - Needs Support	S - Needs Support	Correct
S18	S - Needs Support	S - Needs Support	Correct

Number correct: 12 Number of errors: 6

$$\text{Accuracy} = \frac{\text{exact matches}}{\text{total predictions}} = \frac{12}{18} = 0.6667 = 66.7\%$$

Accuracy tells how often the complete label was correct. It cannot show whether the model mainly confuses Ready with Practice or misses students who Need Support.

★ WHY ACCURACY IS NOT ENOUGH

Accuracy counts every wrong label as one error, but it does not identify the **type or direction** of that error. Consider two very different kinds of mistakes:

- **Error Type 1 — Missed Ready:** A student who is actually *Ready* but the model predicts *Needs Practice* or *Needs Support*. The student was capable, yet the model failed to recognise this.
- **Error Type 2 — False Ready:** A student who actually *Needs Practice* or *Needs Support* but the model predicts *Ready*. The student needed help, yet the model labelled them as ready.

Both error types reduce accuracy by exactly one count. The accuracy score by itself **does not distinguish between these different kinds of incorrect predictions**. It cannot tell you whether the model is primarily missing students who need support, or whether it is wrongly labelling struggling students as ready. To understand *what kinds* of mistakes the model is making and *which classes are being confused with one another*, you need additional tools such as the confusion matrix, precision, and recall.

Confusion-matrix convention (provided):

- **Rows are actual classes;** columns are predicted classes (scikit-learn convention).
- Diagonal cells are exact matches; off-diagonal cells are specific confusions.

PRACTICE P6 · READ AND REASON FROM THE CONFUSION MATRIX

The completed matrix is provided. Use it to analyse the error pattern.

Actual / Predicted	Pred R	Pred P	Pred S	Actual total
Actual R - Ready	6	1	1	8
Actual P - Needs Practice	1	4	1	6
Actual S - Needs Support	0	2	2	4
Predicted total	7	7	4	18

A. Mathematical check

$$\text{Accuracy} = \frac{6 + 4 + 2}{18} = \underline{0.667}, \quad \text{off-diagonal total} = \underline{6}.$$

B. Match each object (1–4) to one provided location (A–D).

Object	Location option
1 Number of actual members of a class	A Off-diagonal cell
2 Number of times a class was predicted	B Row total
3 Exact matches	C Column total
4 One specific direction of error	D Main diagonal

Matches: 1 → B 2 → C 3 → D 4 → A

C. True or false.

- (i) The cell (Actual R, Predicted S) counts Ready students labelled Support. True
- (ii) The Predicted-S total of 4 means all four were actually Support. False
- (iii) Accuracy alone identifies every direction of error. False

D. Application. Suppose S15 is corrected from Predicted P to Predicted S. Complete only the quantities that change:

$$(S, P) : 2 \rightarrow \underline{1}, \quad (S, S) : 2 \rightarrow \underline{3}, \quad \text{Accuracy} : \frac{12}{18} \rightarrow \frac{13}{18} = \underline{72.2\%}.$$

Predicted-P total: 7 → 6; Predicted-S total: 4 → 5; actual row totals do not change.

DO NOT LET THIS MISCONCEPTION SURVIVE

Always state the matrix orientation. Transposing rows and columns changes which real-world error each off-diagonal cell represents.

SECTION 8 Needs-Support Focus: Precision, Recall, and F1**▷ CENTRAL QUESTION**

For this focused analysis, make Needs Support the positive class and group Ready and Needs Practice as “Not Support.” Which predictions become TP, FP, FN, and TN?

★ DEFINE THE BINARY VIEW FIRST

Positive means the class currently being evaluated (here, Needs Support); it does not mean “good.” **Negative** means all other classes grouped together (Ready or Needs Practice); it does not mean “bad.” **True** means correct and **false** means incorrect in this temporary binary view.

Term	Actual status	Predicted status	Meaning here
TP	Support	Support	Correct Support alert
FN	Support	Not Support	Support student missed
FP	Not Support	Support	False Support alert
TN	Not Support	Not Support	Support label correctly rejected

PRACTICE P7 · NEEDS-SUPPORT BINARY VIEW

A. Match each student to TP, FN, FP, or TN in this focused view.

Student	Actual → predicted	Binary category
S17	Support → Support	TP
S15	Support → Practice	FN
S08	Ready → Support	FP
S07	Ready → Practice	TN

B. Total all 18 students in the collapsed table.

Actual / Predicted	Pred Support	Pred Not Support	Actual total
Actual Needs Support	TP = 2	FN = 2	4
Actual Not Support (R or P)	FP = 2	TN = 12	14
Predicted total	4	14	18

★ IMPORTANT MULTICLASS NUANCE

A Ready → Practice prediction is wrong in the full three-class task, yet it is TN in the temporary binary view because the Support label was correctly rejected. TN completes the binary confusion view, although precision, recall, and F1 below do not use it.

Metric map (definitions and formulas are provided). In the F1 row, Prec means precision and Rec means recall.

Metric	Formula	Meaning for Needs Support
Precision	$\frac{TP}{TP + FP}$	Among predicted Support alerts, how many are correct?
Recall / TPR	$\frac{TP}{TP + FN}$	Among actual Support students, how many are found?
F1	$\frac{2TP}{2TP + FP + FN} = \frac{2TP}{\text{Prec} + \text{Rec}}$	Harmonic balance of precision and recall; high only when both are high.

PRACTICE P8 · COMPUTE NEEDS-SUPPORT METRICS

$$\text{Precision}_S = \frac{2}{4} = \underline{0.500},$$

$$\text{Recall}_S = \text{TPR}_S = \frac{2}{4} = \underline{0.500},$$

$$F1_S = \frac{2(\underline{2})}{2(\underline{2}) + \underline{2} + \underline{2}} = \underline{0.500}.$$

Result snapshot (provided):

Denominator group	Counts	Result
4 predicted Support	2 correct; 2 false alerts	Precision = 0.500
4 actual Support	2 found; 2 missed	Recall / TPR = 0.500
Precision and recall	Both are 0.500	F1 = 0.500

★ KEEP THE DENOMINATORS STRAIGHT

- Precision uses predicted positives: $TP + FP$.
- Recall / TPR uses actual positives: $TP + FN$.
- F1 balances precision and recall and does not use TN. TN still completes the binary confusion view.

PRACTICE P9 · WHAT CHANGES WHEN ONE MISS IS CORRECTED?

Suppose S15 changes from Predicted Practice to Predicted Support. The updated counts are $TP = 3$, $FP = 2$, $FN = 1$, and $TN = 12$. Complete the three core metrics.

$$\text{Precision}_S = \frac{3}{5} = \underline{0.600}, \quad \text{Recall}_S = \frac{3}{4} = \underline{0.750},$$

$$F1_S = \frac{2(\underline{3})}{2(\underline{3}) + \underline{2} + \underline{1}} = \underline{0.667}.$$

- (a) If missing Needs Support students is costliest, prioritise Recall / TPR.
- (b) If unnecessary Support alerts are costliest, prioritise Precision.
- (c) If both concerns matter, use F1.

★ THE IMPORTANT PATTERN

Overall accuracy is 0.667, but the model finds only 2 of 4 actual Needs Support students. Its Support precision, recall, and F1 are all 0.500. Overall correctness can hide weak performance on a smaller, operationally important class.

Undefined cases: if a model never predicts a class, its precision denominator is zero. If the evaluation set contains no actual examples of a class, its recall denominator is zero. A report must state how the library handled the undefined value.

SECTION 9 Macro, Weighted, and Micro Averages**▷ CENTRAL QUESTION**

Accuracy gives one overall count. When precision, recall, or F1 must be summarised across several classes, should every class receive equal weight, or should larger classes contribute more?

★ PROVIDED INPUTS FOR AGGREGATION

Return to the **original Master Test Data C**; Practice P9 was a what-if only. Use these supplied class summaries; students are not expected to recompute them here.

Class	Actual count	Precision	Recall	F1
Ready	8	0.857	0.750	0.800
Needs Practice	6	0.571	0.667	0.615
Needs Support	4	0.500	0.500	0.500

Macro average - every class gets one equal vote.

$$\text{Macro}(M) = \frac{M_R + M_P + M_S}{3}$$

$$\text{Macro Precision} = \frac{0.857 + 0.571 + 0.500}{3} = \underline{0.643},$$

$$\text{Macro Recall} = \frac{0.750 + 0.667 + 0.500}{3} = \underline{0.639},$$

$$\text{Macro F1} = \frac{0.800 + 0.615 + 0.500}{3} = \underline{0.638}.$$

Macro averaging ignores class size. Use it when a smaller class should matter as much as a larger class.

Weighted average - use actual class count as the weight.

$$\text{Weighted}(M) = \frac{8M_R + 6M_P + 4M_S}{18}$$

$$\text{Weighted Precision} = \frac{8(6/7) + 6(4/7) + 4(1/2)}{18} = \frac{43}{63} \approx \underline{\underline{0.683}},$$

$$\text{Weighted Recall} = \frac{8(0.750) + 6(0.667) + 4(0.500)}{18} \approx \underline{\underline{0.667}},$$

$$\text{Weighted F1} = \frac{8(0.800) + 6(0.615) + 4(0.500)}{18} \approx \underline{\underline{0.672}}.$$

Micro average - pool all class decisions first: $TP = 12$, $FP = 6$, $FN = 6$.

$$\text{Micro Precision} = \frac{12}{12 + 6} = \underline{\underline{0.667}},$$

$$\text{Micro Recall} = \frac{12}{12 + 6} = \underline{\underline{0.667}},$$

$$\text{Micro F1} = \frac{2(12)}{2(12) + 6 + 6} = \underline{\underline{0.667}}.$$

Every wrong single-label prediction contributes one pooled FP for the predicted class and one pooled FN for the missed actual class. Therefore micro precision, recall, and F1 equal accuracy in this task.

PRACTICE P10 · CHOOSE THE AVERAGING RULE

Summary	Precision	Recall	F1 / score	Interpretation
Accuracy	—	—	0.667	Exact-label correctness
Macro	0.643	0.639	0.638	Treat all classes equally
Weighted	0.683	0.667	0.672	Reflect actual class proportions
Micro	0.667	0.667	0.667	Pool decisions; equals accuracy here

Why is weighted F1 higher than macro F1 here?

Answer: Needs Support is the weakest class but contains only 4 of 18 students. Weighted averaging gives it less influence than macro averaging, so the weighted score is higher.

DO NOT LET THIS MISCONCEPTION SURVIVE

Macro F1 is the arithmetic average of the class F1 scores. It is generally **not** the harmonic mean of macro precision and macro recall.

SECTION 10 Metric Choice and Final Synthesis

▷ CENTRAL QUESTION

Do not begin with the formula or the score that looks largest. Begin with the decision: which error is costly, and what information is needed to act?

PRACTICE P11 · MATCH THE DECISION TO THE METRIC

Match each **decision need** in Column A with the most useful **metric** / **diagnostic** from Column B. Write the letter in the space provided.

Column B — Metric / Diagnostic Options

Option	Metric / Diagnostic	Option	Metric / Diagnostic
(a)	MAPE	(g)	Confusion matrix
(b)	Recall / TPR	(h)	Accuracy (or micro F1 here)
(c)	Adjusted R^2 plus validation	(i)	MAE
(d)	Weighted average	(j)	RMSE or MSE
(e)	F1	(k)	R^2 with MAE/RMSE
(f)	Precision	(l)	Macro average

Column A — Decision Needs

Decision need	Match (letter)
1 Typical regression error in stipend units	(i) MAE
2 Large regression misses deserve extra weight	(j) RMSE/MSE
3 Relative regression error is meaningful and actuals stay away from zero	(a) MAPE
4 Compare fit with the mean-only regression baseline	(k) R^2
5 Compare same-data linear models with different predictor counts	(c) Adj. R^2
6 All exact-label mistakes have similar cost	(h) Accuracy
7 See exactly which class labels are confused	(g) Conf. matrix
8 Missing actual Needs Support students is costly	(b) Recall
9 False Needs Support alerts are costly	(f) Precision
10 Balance Support precision and recall	(e) F1
11 Give every class equal importance	(l) Macro avg
12 Let actual class proportions determine weight	(d) Weighted avg

◆ TAKEAWAY

A model is not good because one score is high. It is good when its errors are acceptable for the decision we need to make, on data that represents the world in which the model will be used.

EXIT CHECK · THREE QUESTIONS

- How can two regression datasets have the same RMSE but different R^2 values?

Answer: They can have the same RSS/RMSE but different target spread and therefore different TSS.

- Why did Model B obtain a higher R^2 but a lower Adjusted R^2 than Model A?

Answer: Its lower RSS raised raw R^2 , but the small gain did not overcome the larger penalty for a third predictor.

- How can the classifier be 66.7% accurate while recall for Needs Support is only 50%?

Answer: Accuracy counts 12 correct labels across all 18 students. Support recall considers only the four actual Support students, of whom the model found two. Overall correctness can hide minority-class misses.

★ BY THE END OF THIS SHEET, YOU CAN

- Explain the difference between training loss and evaluation metrics.
- Interpret MAE, MAPE, MSE, RMSE, R^2 , and Adjusted R^2 on one regression dataset.
- Read a multiclass confusion matrix with rows as actual and columns as predicted.
- Calculate TP, FP, FN, TN, precision, recall/TPR, and F1 for the Needs-Support focus.
- Compare supplied class scores using macro, weighted, and micro aggregation.
- Choose a metric from the decision cost rather than from the largest numerical score.